

Root cause analysis in IT monitoring systems

Charles Assaad



- 1 Overview
 - IT monitoring systems
 - Causal graphs and abstractions
 - Root causes and anomalies
- 2 Root cause analysis using causal graphs
 - A root cause analysis framework using causal discovery
 - New root cause analysis framework
- 3 Experiments
- 4 Conclusion

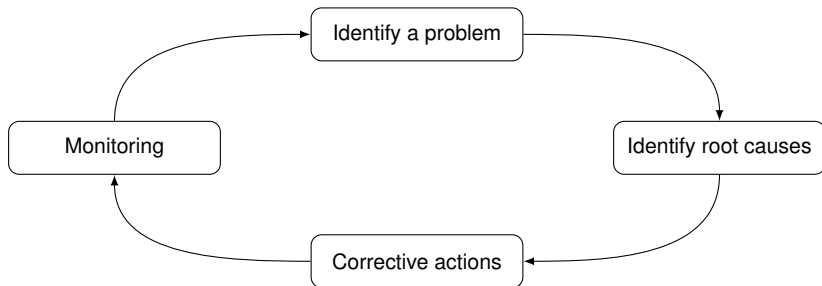
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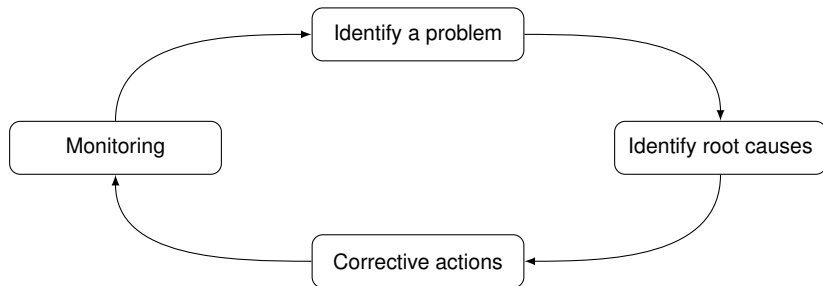
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IT monitoring system

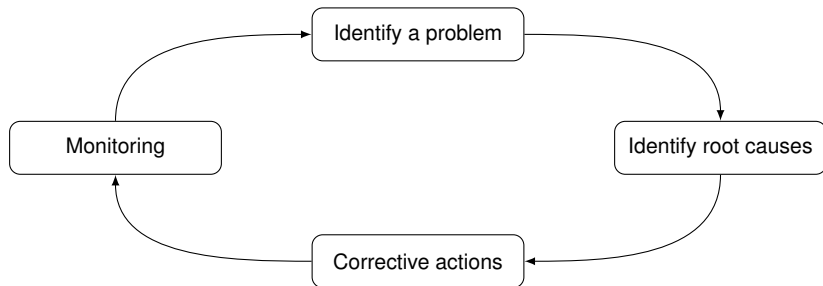


IT monitoring system



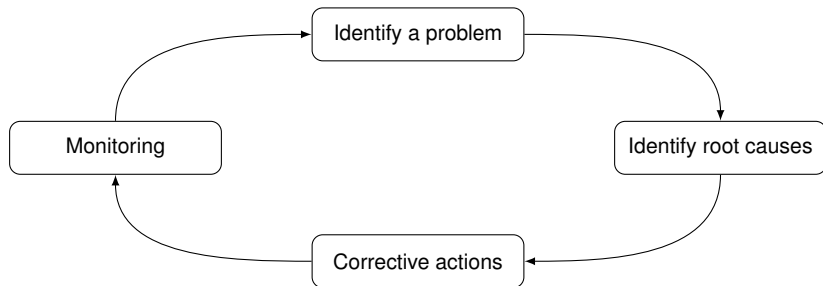
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IT monitoring system



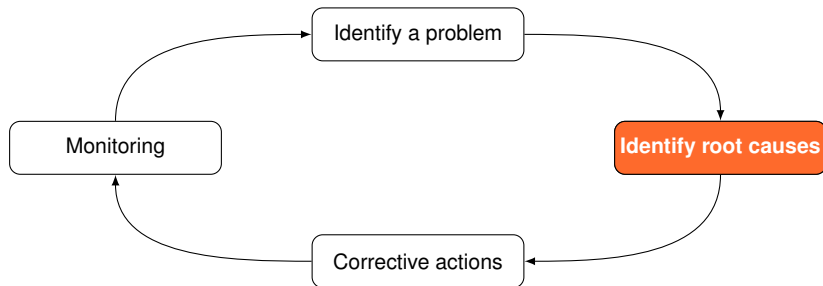
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IT monitoring system



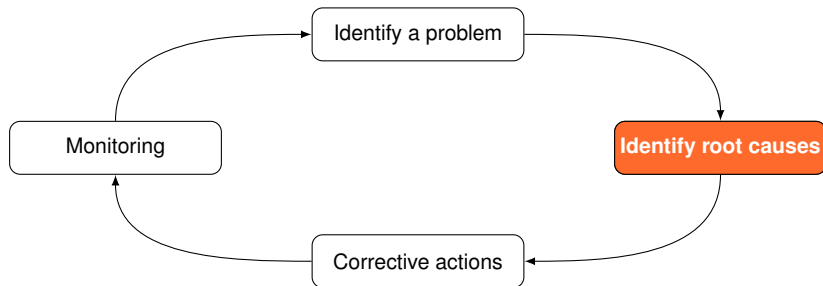
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- Corrective actions $\Rightarrow$ costly interventions

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IT monitoring system



- Identify a problem $\Rightarrow$ anomaly detection
- Identify causes $\Rightarrow$ **root cause analysis**
- Corrective actions $\Rightarrow$ costly interventions

Challenges

- Time series data
- Interaction with system experts
- Need for high availability
 - Low time computation
 - Small size of anomalous time series
 - No labeled data

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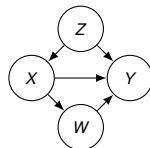
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Causal graph

- $\mathcal{G}$: directed acyclic graph (DAG)
- $\mathcal{V}$: set of vertices in $\mathcal{G}$
- $\mathcal{E}$: set of arcs in $\mathcal{G}$
- d : number of vertices in $\mathcal{G}$

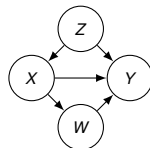
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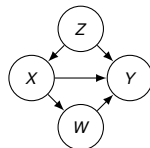
Causal Markov condition (CMC)

$$\forall X \in \mathcal{V}, X \perp\!\!\!\perp_{\mathcal{P}} \mathcal{V} \setminus \{ \text{Descendants}(X), \text{Parents}(X) \} \mid \text{Parents}(X)$$

- $\mathcal{P}$: distribution compatible with the graph $\mathcal{G}$
- $\perp\!\!\!\perp_{\mathcal{P}}$: Statistical independence

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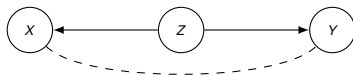
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- $\mathcal{P}$: distribution compatible with the graph $\mathcal{G}$
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Causal sufficiency

$$\forall X, Y \in \mathcal{V}, \forall Z, X \leftarrow Z \rightarrow Y \implies Z \in \mathcal{V}$$

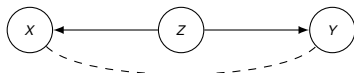
Blocked path



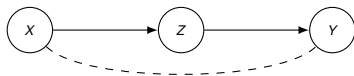
Fork



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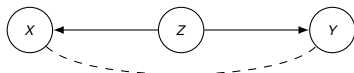


Chain

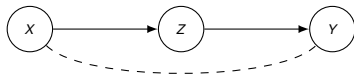


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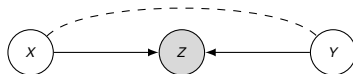
Fork



Chain

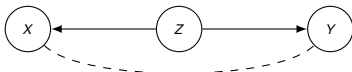


V-structure

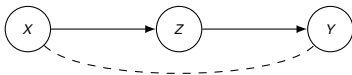


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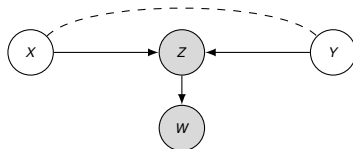
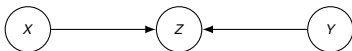
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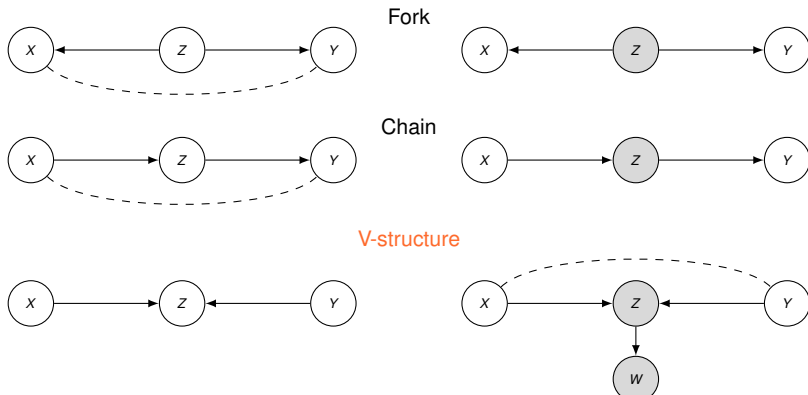
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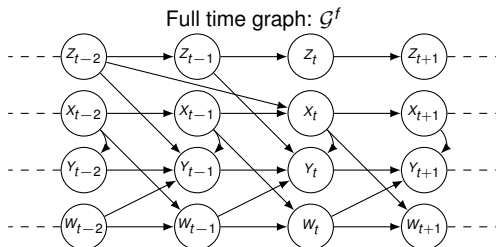


Definition (Blocked path)

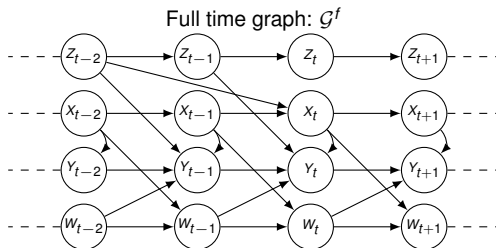
A path between two vertices X and Y is blocked by a conditioning set $S \subset \mathcal{V}$ if either of the following is true:

- there is a fork $\dots \leftarrow Z \rightarrow \dots$ or a chain $\dots \rightarrow Z \rightarrow \dots$ on the path such that $Z \in S$;
- there is a V-structure $\dots \rightarrow Z \leftarrow \dots$ on the path such that $Z \notin S$ and $\text{Descendants}(Z) \notin S$.

Temporal causal graphs and abstractions



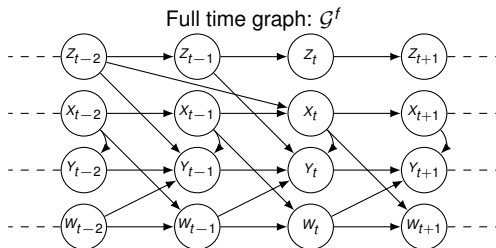
Temporal causal graphs and abstractions



Additional assumption: stationarity

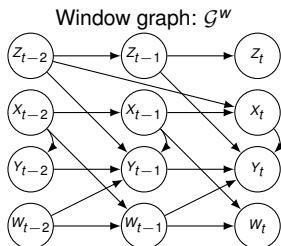
$\gamma_{\max}$: maximal lag between causes and their effects

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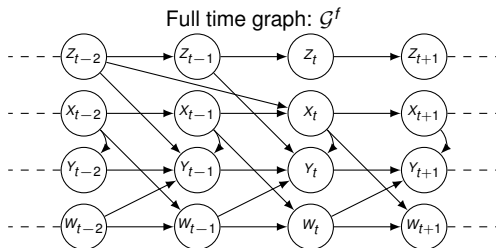


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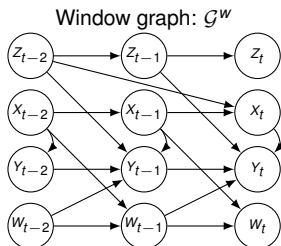


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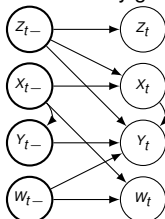


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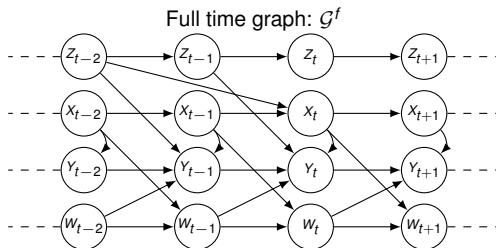
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Extended summary graph: $\mathcal{G}^e$

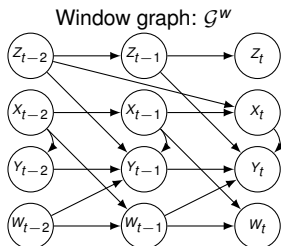


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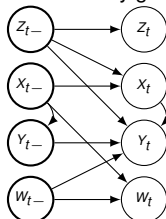


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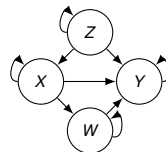
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Extended summary graph: $\mathcal{G}^e$



Summary graph (SCG): $\mathcal{G}^s$



Outline

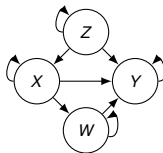
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Root causes and anomalies

- Root causes are vertices in the graph that underwent an external intervention

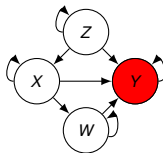
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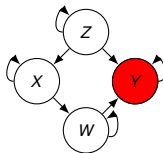
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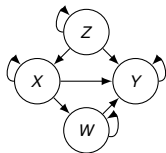
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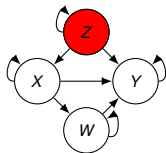
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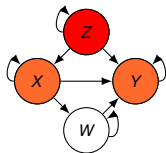
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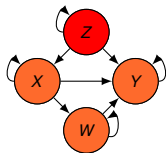
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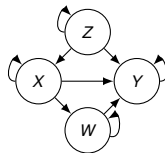
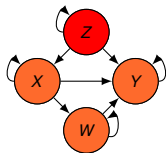
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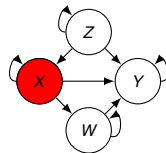
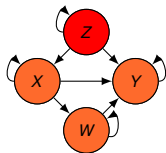
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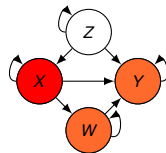
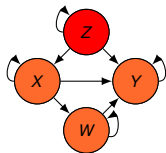
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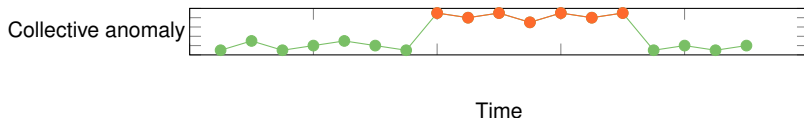
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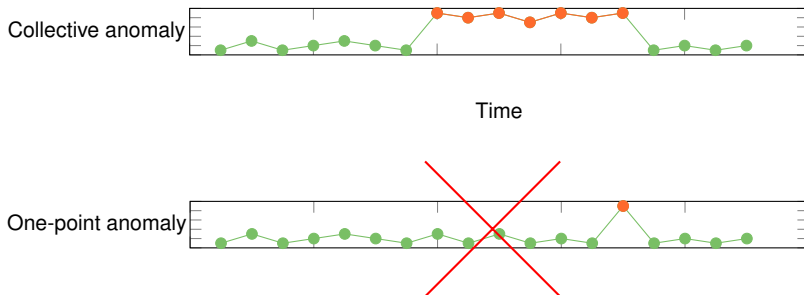
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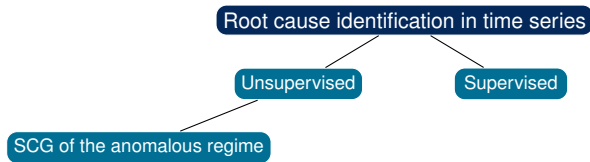
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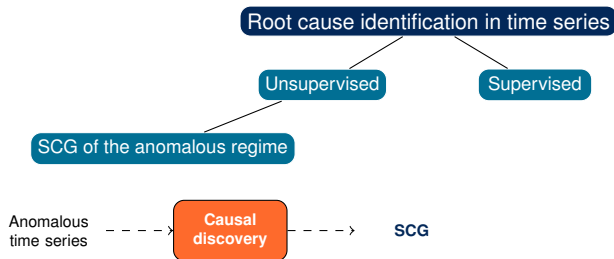
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Root cause analysis using causal discovery

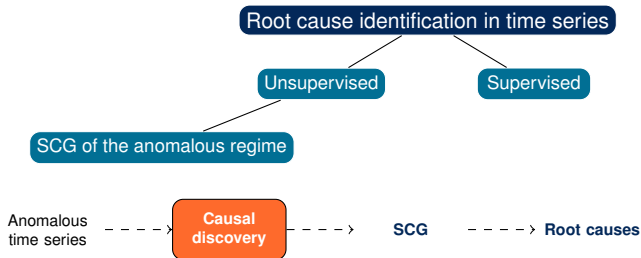


Root cause analysis using causal discovery



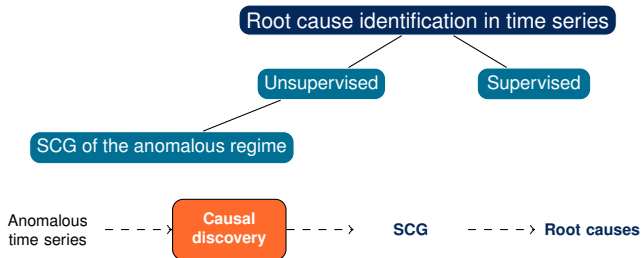
- Discover the SCG from anomalous time series

Root cause analysis using causal discovery



- Discover the SCG from anomalous time series
- All anomalous root vertices in the inferred graph are considered as root causes

Root cause analysis using causal discovery

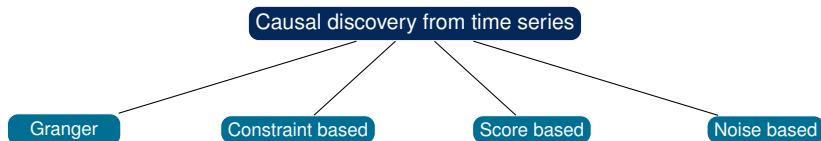


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Limitations:

- In theory, these methods only consider structural interventions
- They require additional assumptions (depending on the causal discovery algorithm they use)

Causal discovery from time series



A. Arnold et al. *Temporal causal modeling with graphical granger method*. KDD, 2007.

J. Runge. *Discovering contemporaneous and lagged causal relations in autocorrelated nonlinear time series datasets*. UAI, 2020.

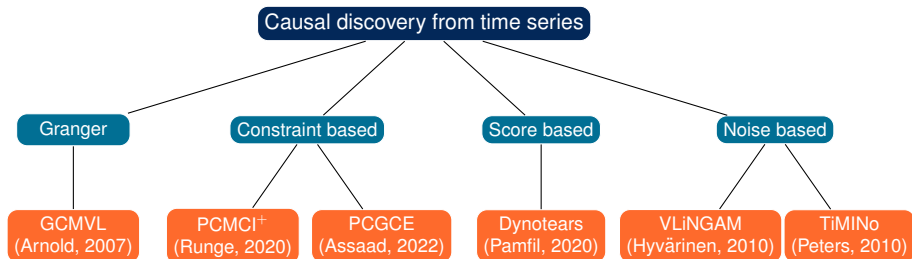
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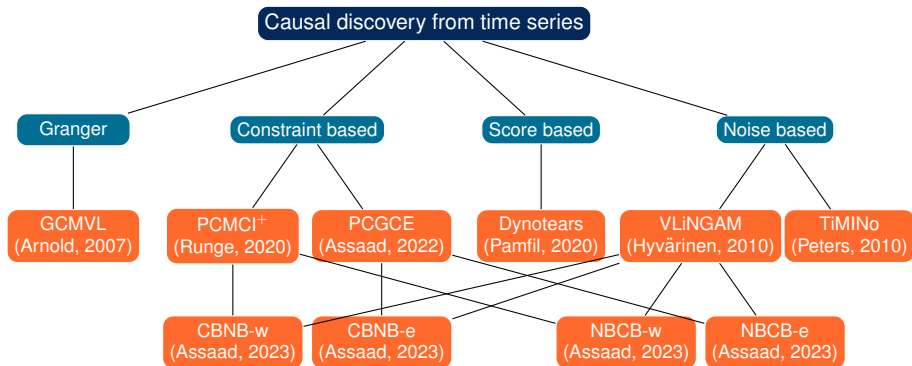
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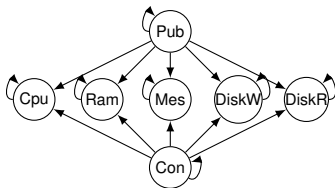
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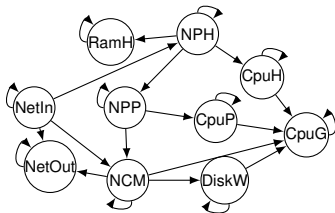
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IT monitoring case studies



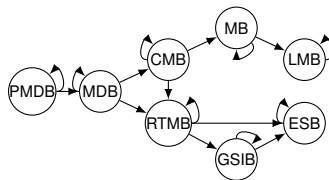
Message ingestion activity based on Publish/Subscribe architecture

Two datasets of 288 and 364 timestamps



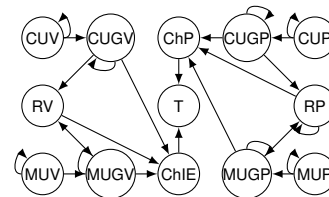
Web activity

Two datasets of 3000 timestamps



Message ingestion activity based on Storm architecture

One dataset of 991 timestamps



Antivirus activity

Two datasets of 1321 timestamps

IT monitoring case studies: results

Estimation

- Partial correlation
- ...

Hyper-parameters

- $\gamma_{max} = 3$
- Significance level = 0.05
- ...

Evaluation measure

- F1-score on orientations in the SCG.

IT monitoring case studies: results

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Results

	MoM 1	MoM 2	Ingestion	Web 1	Web 2	Antivirus 1	Antivirus 2
GCMVL	0.0	0.0	0.2	0.2	0.0	0.08	0.0
Dynotears	0.26	0.2	0.14	0.23	0.3	0.18	0.19
PCMCI ⁺	0.4	0.0	0.0	0.23	0.3	0.04	0.11
PCGCE	0.0	0.12	0.12	0.22	0.15	0.3	0.45
VLiNGAM	0.0	0.0	0.19	0.29	0.18	0.15	0.22
TiMINo	0.0	0.17	0.18	0.0	0.0	0.0	0.0
NBCB-w	0.4	0.0	0.13	0.23	0.3	0.14	0.24
NBCB-e	0.13	0.29	0.27	0.19	0.42	0.31	0.45
CBNB-w	0.4	0.0	0.15	0.23	0.3	0.17	0.16
CBNB-e	0	0.24	0.13	0.22	0.29	0.31	0.38

IT monitoring case studies: results

Estimation

- Partial correlation
- ...

Hyper-parameters

- $\gamma_{max} = 3$
- Significance level = 0.05
- ...

Evaluation measure

- F1-score on orientations in the SCG.

Results

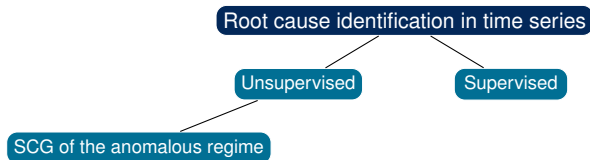
	MoM 1	MoM 2	Ingestion	Web 1	Web 2	Antivirus 1	Antivirus 2
GCMVL	0.0	0.0	0.2	0.2	0.0	0.08	0.0
Dynotears	0.26	0.2	0.14	0.23	0.3	0.18	0.19
PCMCI ⁺	0.4	0.0	0.0	0.23	0.3	0.04	0.11
PCGCE	0.0	0.12	0.12	0.22	0.15	0.3	0.45
VLiNGAM	0.0	0.0	0.19	0.29	0.18	0.15	0.22
TiMINo	0.0	0.17	0.18	0.0	0.0	0.0	0.0
NBCB-w	0.4	0.0	0.13	0.23	0.3	0.14	0.24
NBCB-e	0.13	0.29	0.27	0.19	0.42	0.31	0.45
CBNB-w	0.4	0.0	0.15	0.23	0.3	0.17	0.16
CBNB-e	0	0.24	0.13	0.22	0.29	0.31	0.38

Not satisfactory for real applications!

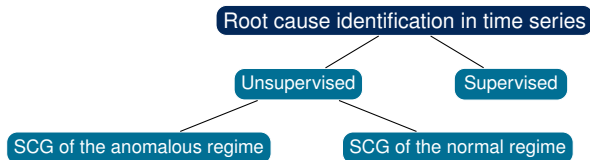
Outline

- 1 Overview
 - IT monitoring systems
 - Causal graphs and abstractions
 - Root causes and anomalies
- 2 Root cause analysis using causal graphs**
 - A root cause analysis framework using causal discovery
 - New root cause analysis framework**
- 3 Experiments
- 4 Conclusion

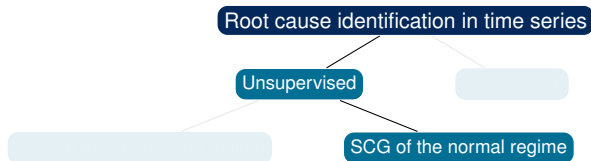
New root cause analysis framework



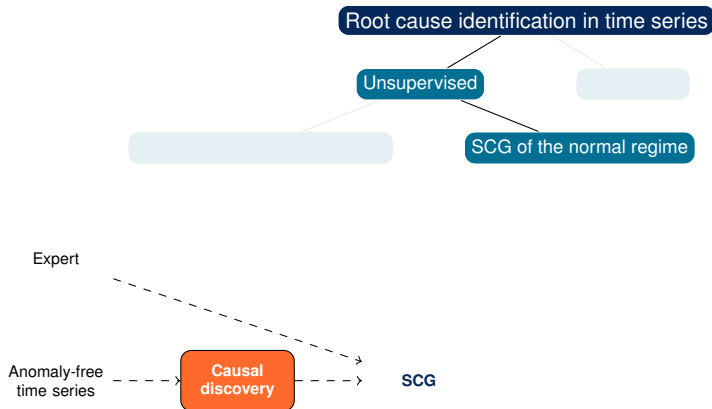
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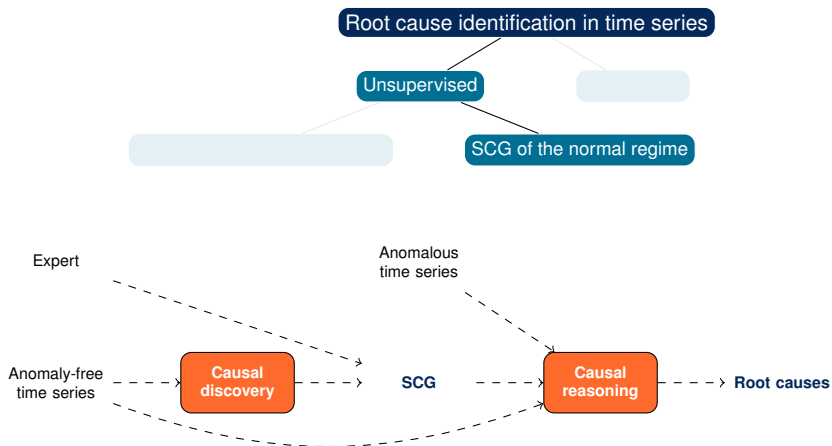
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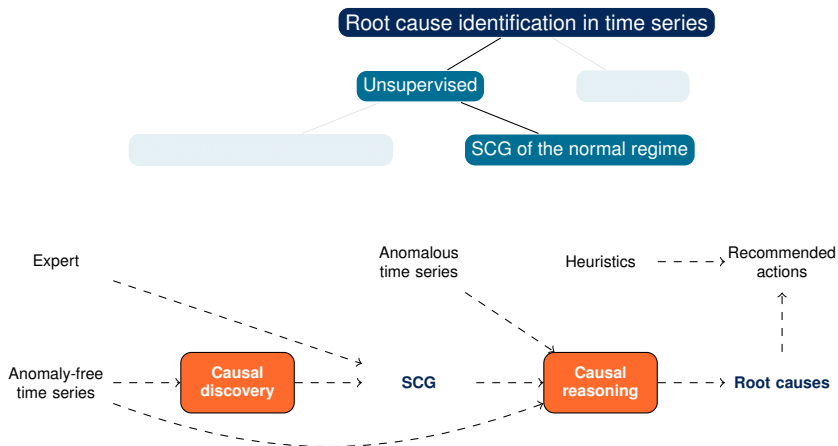
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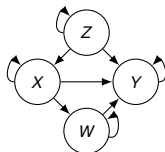


Additional assumption on SCGs

- Acyclic SCG with loops (ASCGL)

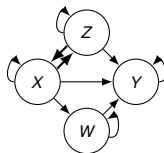
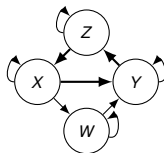
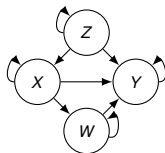
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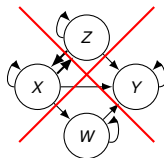
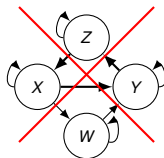
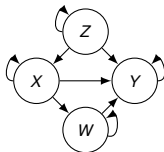
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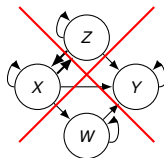
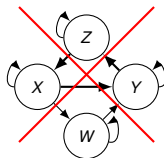
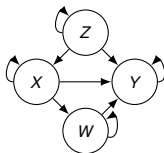
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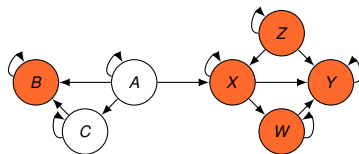
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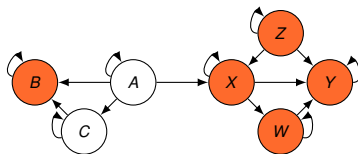


This assumption will be relaxed later

Grouping related anomalies



Grouping related anomalies

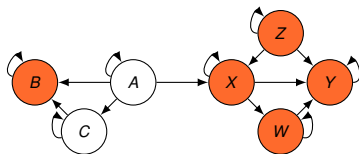


Definition (Linked anomalous graph)

A linked anomalous graph $\mathcal{L}^i$ is a subgraph of $\mathcal{G}^S$ such that

- each vertex in $\mathcal{L}^i$ is anomalous
- all vertices in $\mathcal{G}^S$ that are adjacent to a vertex in $\mathcal{L}^i$ are also in $\mathcal{L}^i$
- all vertices in $\mathcal{L}^i$ are connected with each other through an anomalous path

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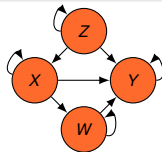


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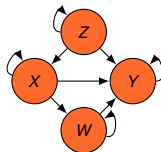
Linked anomalous graph 1



Linked anomalous graph 2

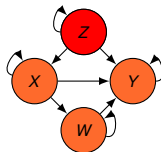


Finding external interventions from graph



C. K. Assaad et al. *Root cause identification for collective anomalies in time series given an acyclic summary causal graph with loops*. AISTATS, 2023

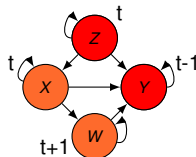
Finding external interventions from graph



Definition (Sub root vertex)

A sub root vertex is root vertex in a $\mathcal{L}^i$.

Finding external interventions from graph



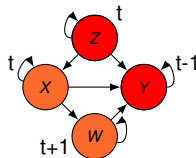
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Definition (Time defying vertex)

A time defying vertex is a vertex in $\mathcal{L}^i$ such that the appearance time of the anomaly on Y precedes the appearance time of the anomaly on each $X \in \text{Parents}(Y, \mathcal{L}^i)$.

Finding external interventions from graph



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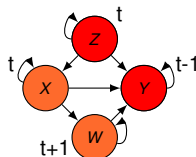
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Sub root vertices + Time defying vertices $\subset$ **Root causes**

Finding external interventions from graph



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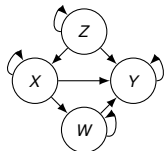
Sub root vertices + Time defying vertices $\subset$ **Root causes**

What about **Root causes** – Sub root vertices – Time defying vertices?

C. K. Assaad et al. *Root cause identification for collective anomalies in time series given an acyclic summary causal graph with loops*. AISTATS, 2023

Finding external interventions from data

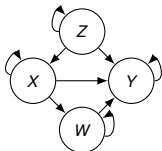
Normal regime



$$\begin{cases} Z_t := \alpha_{Z_{t-1}, Z_t} * Z_{t-1} + \xi_{Z_t} \\ X_t := \alpha_{X_{t-1}, X_t} * X_{t-1} + \alpha_{X_t, Z_{t-2}} * Z_{t-2} + \xi_{X_t} \\ W_t := \alpha_{W_{t-1}, W_t} * W_{t-1} + \alpha_{W_t, X_{t-1}} * X_{t-1} + \xi_{W_t} \\ Y_t := \alpha_{Y_{t-1}, Y_t} * Y_{t-1} + \alpha_{X_t, Y_t} * X_t + \alpha_{W_{t-1}, Y_t} * W_{t-1} + \alpha_{Z_{t-1}, Y_t} * Z_{t-1} + \xi_{Y_t} \end{cases}$$

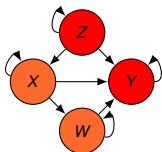
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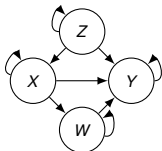
Anomalous regime: Intervention on Z_t and Y_t



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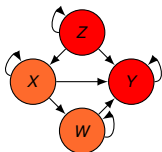
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Definition (Data defying vertex)

A data defying vertex Y is a vertex in $\mathcal{L}^i$ such that $\exists X \in \text{Parents}(Y, \mathcal{G}^s)$ and $\exists \gamma_{xy} \in \{0, \dots, \gamma_{\max}\}$ such that $\alpha_{X_{t-\gamma_{xy}}, Y_t} \neq \bar{\alpha}_{X_{t-\gamma_{xy}}, Y_t}$.

Identifiability of direct effects in ASCGL

How to find data defying vertices?

¹ Identify a direct effect $\iff$ find an adjustment set which makes the direct effect computable from the observed distribution.  

Identifiability of direct effects in ASCGL

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For all Y in $\mathcal{L}^i$, for each $X \in \text{Parents}(Y, \mathcal{G}^s)$ and for each $\gamma_{xy} \in \{0, \dots, \gamma_{\max}\}$:

- Identify¹ and estimate $\alpha_{X_t - \gamma_{xy}, Y_t}$ and $\bar{\alpha}_{X_t - \gamma_{xy}, Y_t}$
- If $\alpha_{X_t - \gamma_{xy}, Y_t} \neq \bar{\alpha}_{X_t - \gamma_{xy}, Y_t}$ then Y is a root cause

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$\bar{\gamma}_{xy}$: lag between the appearance of the anomaly on X and the appearance of the anomaly on Y

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Are direct effects identifiable using an ASCGL?

- Yes!

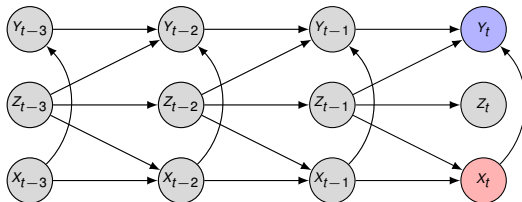
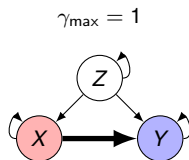
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An adjustment set for ASCGLs

Usually, in DAGs, an adjustment set is a subset of ancestors of Y that block all non-direct paths between X and Y .

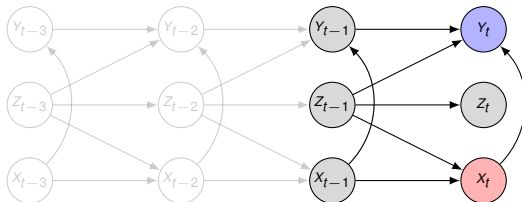
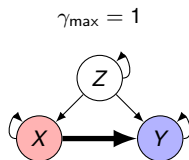
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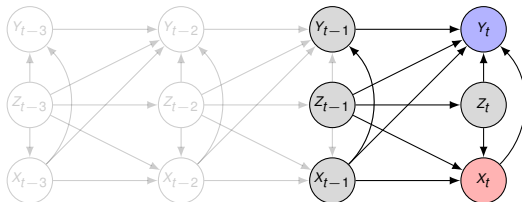
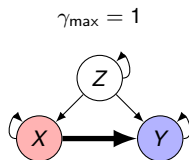
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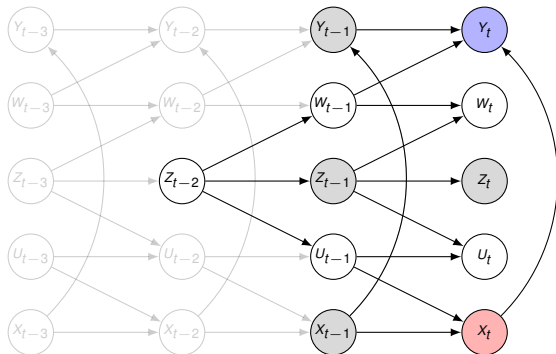
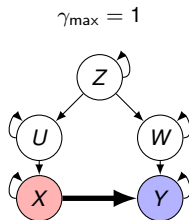
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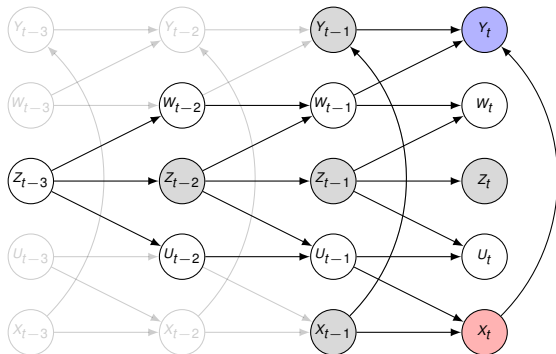
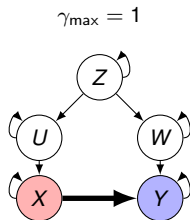
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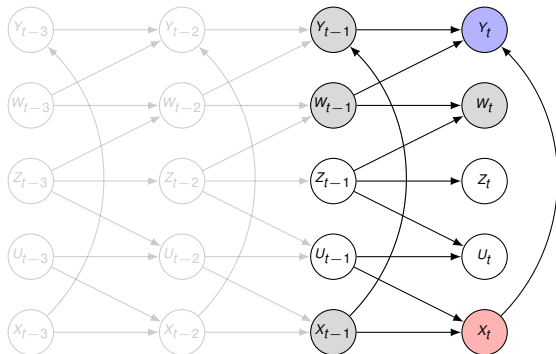
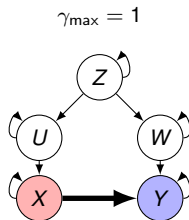
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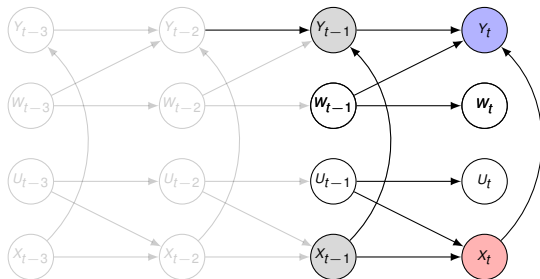
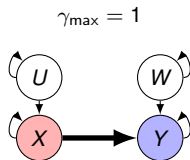
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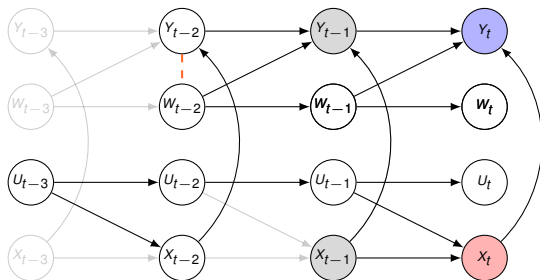
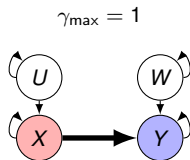
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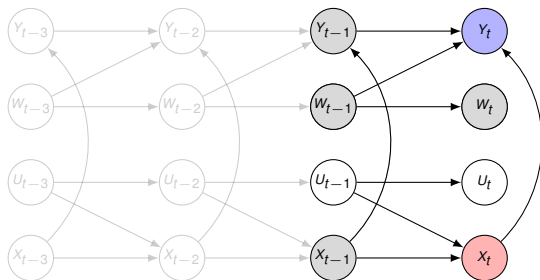
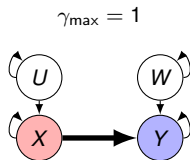
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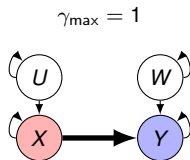
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An adjustment set for ASCGLs

Usually, in DAGs, an adjustment set is a subset of ancestors of Y that block all non-direct paths between X and Y .

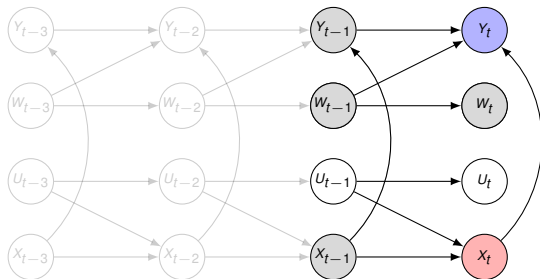


Simple adjustment set:

$$\mathcal{A}_{t-\gamma_{\max}} \cup \dots \cup \mathcal{A}_t \setminus \{X_{t-\gamma_{xy}}, Y_t\}$$

such that

$$\mathcal{A} = \text{Parents}(Y, \mathcal{G}^s)$$



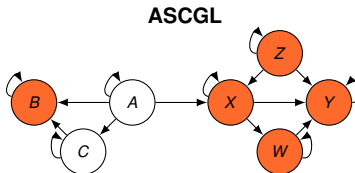
Extention to general SCGs

	ASCGL	SCG
Linked anomalous graph		
Sub root vertex		
Time defying vertex		

S. Ferreira and C. K. Assaad. *Challenges of root Cause Identification for Collective Anomalies in Time Series given a Summary Causal Graph*. Colloquium on When Causal Inference meets Statistical Analysis, 2023

Extention to general SCGs

Linked
anomalous
graph



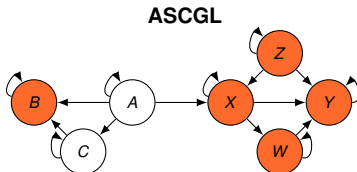
Sub root
vertex

Time
defying
vertex

SCG

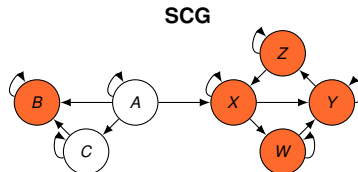
Extention to general SCGs

Linked
anomalous
graph



Sub root
vertex

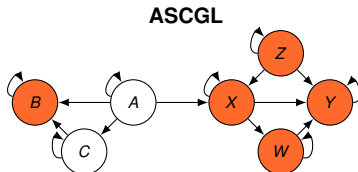
Time
defying
vertex



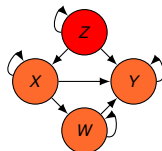
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Extention to general SCGs

Linked
anomalous
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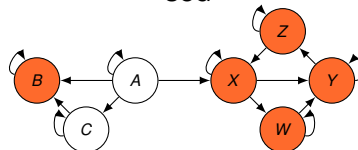


Sub root
vertex



Time
defying
vertex

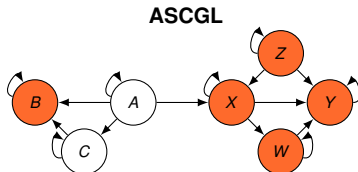
SCG



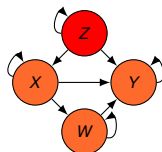
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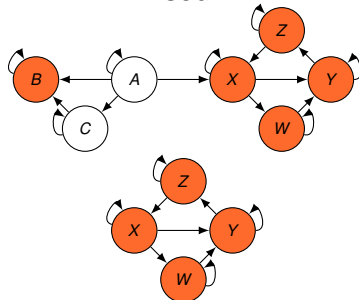


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vertex



Time
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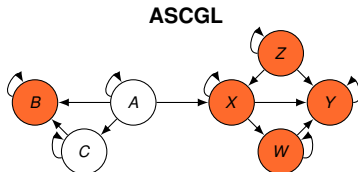
SCG



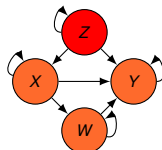
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Extention to general SCGs

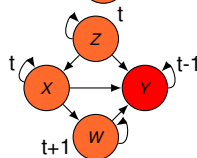
Linked
anomalous
graph



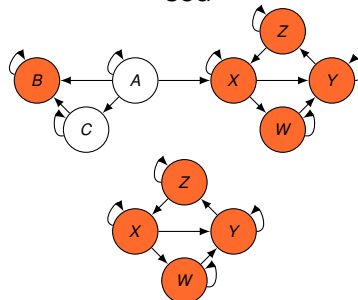
Sub root
vertex



Time
defying
vertex

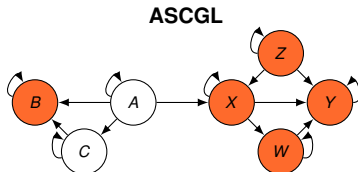


SCG

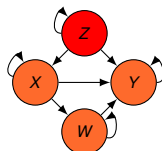


Extention to general SCGs

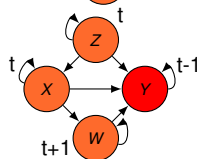
Linked
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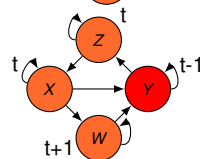
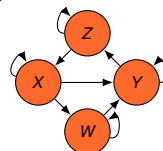
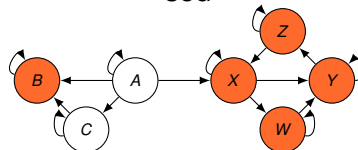
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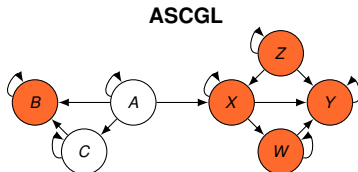


SCG

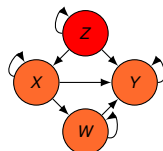


Extention to general SCGs

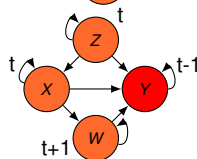
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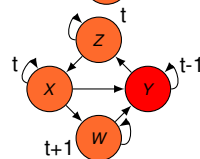
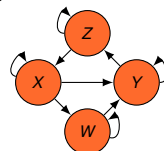
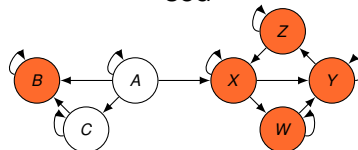
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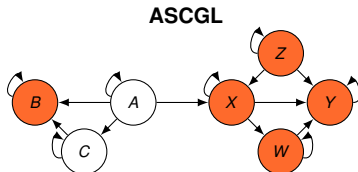
SCG



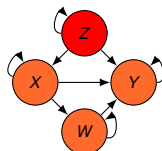
What about data defying vertices?

Extention to general SCGs

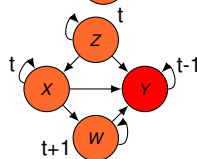
Linked
anomalous
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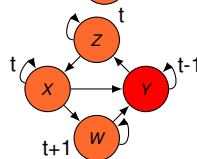
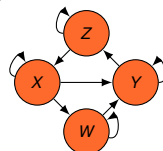
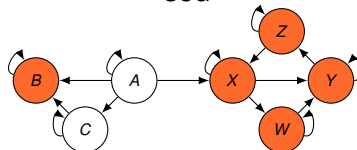
Sub root
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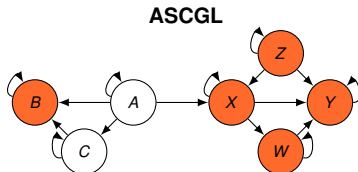
SCG



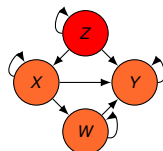
What about data defying vertices? Are direct effects identifiable using a SCG?

Extention to general SCGs

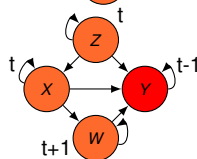
Linked
anomalous
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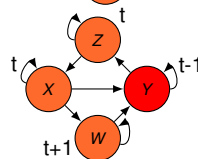
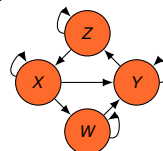
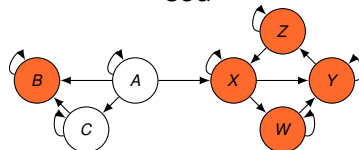
Sub root
vertex



Time
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SCG



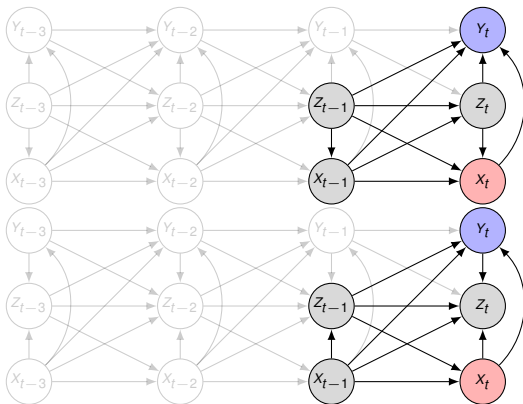
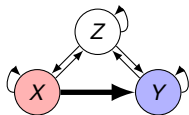
What about data defying vertices? Are direct effects identifiable using a SCG?

- No! Not always!

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Examples

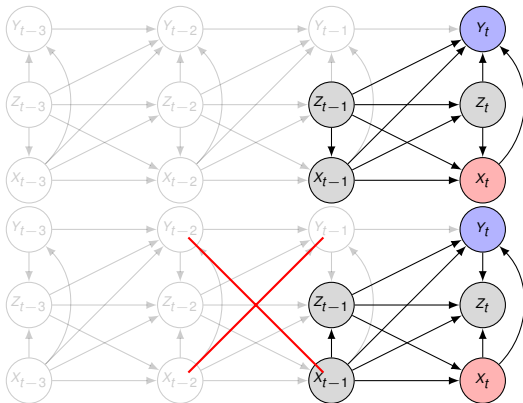
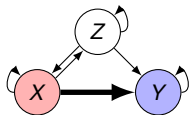
$$\gamma_{\max} = 1$$



Not identifiable

Examples

$$\gamma_{\max} = 1$$



Identifiable

Identifiability of direct effects in SCGs

Theorem

$\alpha_{X_t - \gamma_{xy}, Y_t}$ is not identifiable **if and only if** $X \in \text{Par}(Y, \mathcal{G}^s)$ and one of the following conditions holds:

Identifiability of direct effects in SCGs

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$\alpha_{X_t - \gamma_{xy}, Y_t}$ is not identifiable **if and only if** $X \in \text{Par}(Y, \mathcal{G}^s)$ and one of the following conditions holds:

- 1 $X \in \text{Desc}(Y, \mathcal{G}^s)$ and $\exists C \in \text{Cycles}(X, \mathcal{G}^s)$ with $Y \notin C$, or

Identifiability of direct effects in SCGs

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- ① $X \in \text{Desc}(Y, \mathcal{G}^s)$ and $\exists C \in \text{Cycles}(X, \mathcal{G}^s)$ with $Y \notin C$, or
- ② There exists an active non-direct path $\pi_s = \langle V^1, \dots, V^n \rangle$ from X to Y in $\mathcal{G}^s$ such that $\langle V^2, \dots, V^{n-1} \rangle \subseteq \text{Desc}(Y, \mathcal{G}^s)$, and one of the following conditions holds:
 - ① $\gamma_{xy} = 0$, or
 - ② $\gamma_{xy} > 0$ and $\nexists 1 \leq i < n, V^i \leftarrow V^{i+1}$.

Identifiability of direct effects in SCGs

Theorem

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This is a complete result: it determines if a direct effect is identifiable for every direct effect of every SCG.

An adjustment set for SCGs

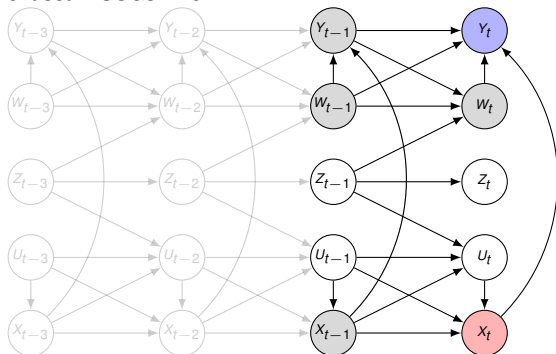
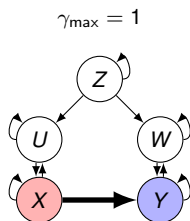
Is the set of parents a sufficient adjustment set in SCGs?

An adjustment set for SCGs

Is the set of parents a sufficient adjustment set in SCGs? No

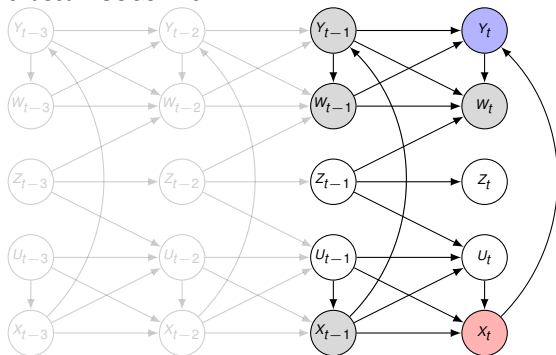
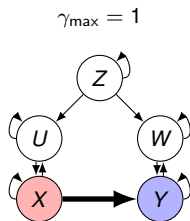
An adjustment set for SCGs

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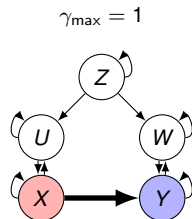
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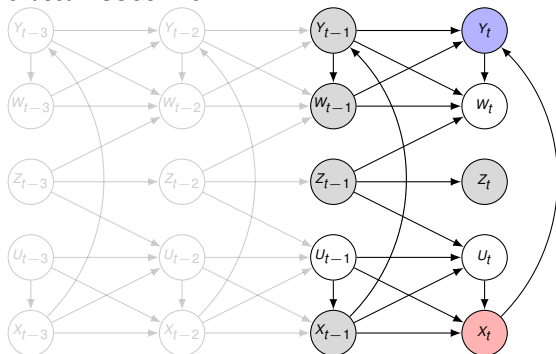


An adjustment set for SCGs

Is the set of parents a sufficient adjustment set in SCGs? No



Simple adjustment set:



$$\mathcal{D}_{t-\gamma_{\max}} \cup \dots \cup \mathcal{D}_{t-1} \cup \mathcal{A}_{t-\gamma_{\max}} \cup \dots \cup \mathcal{A}_t \setminus \{X_{t-\gamma_{xy}}, Y_t\}$$

such that

$$\mathcal{D} = \text{Ancestors}(Y, \mathcal{G}^s) \cap \text{Descendants}(Y, \mathcal{G}^s)$$

$$\mathcal{A} = \text{Ancestors}(Y, \mathcal{G}^s) \setminus \text{Descendants}(Y, \mathcal{G}^s)$$

EasyRCA

Lags between
anomalies

SCG

Anomalous
time series

Anomaly-free
time series

EasyRCA

Lags between
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SCG

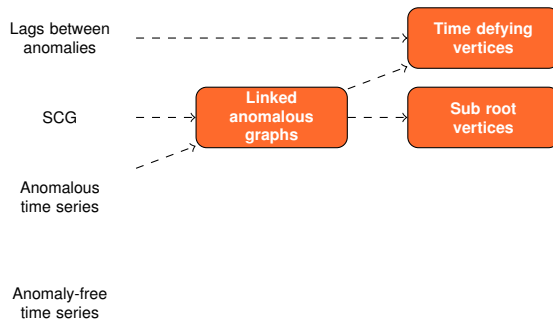


Anomalous
time series

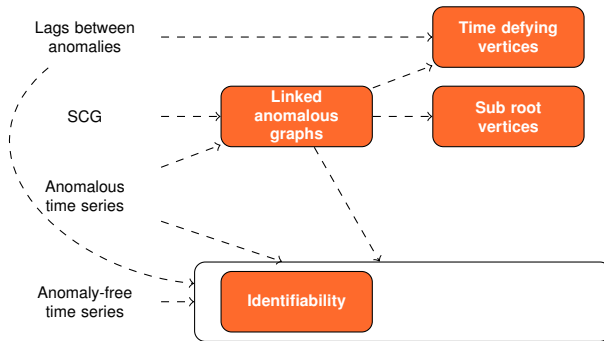
Anomaly-free
time series

**Linked
anomalous
graphs**

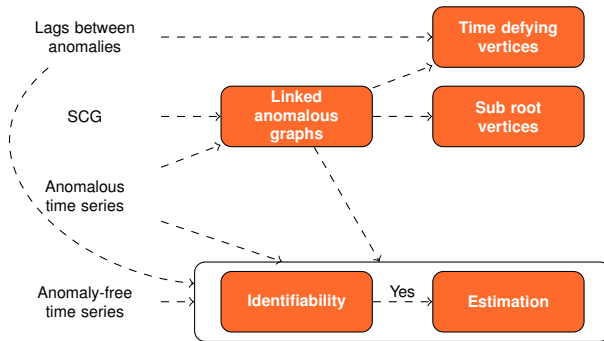
EasyRCA



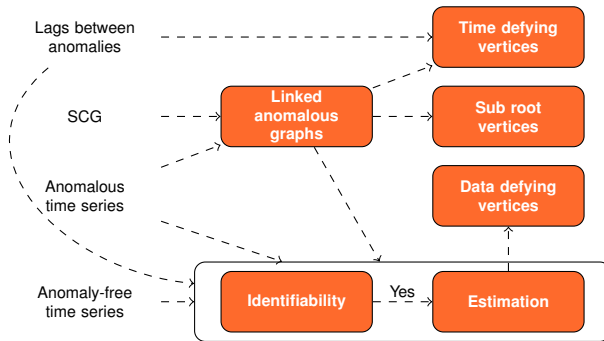
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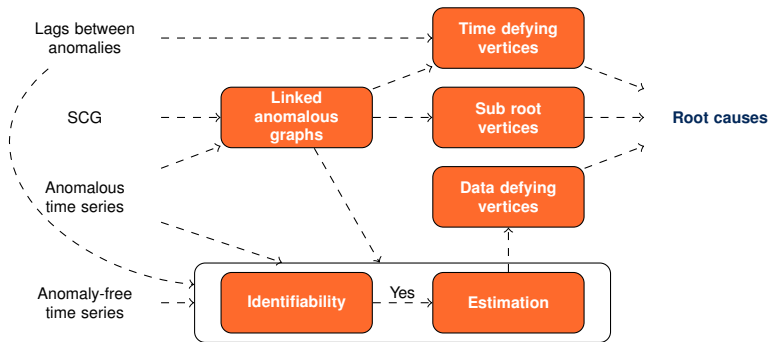
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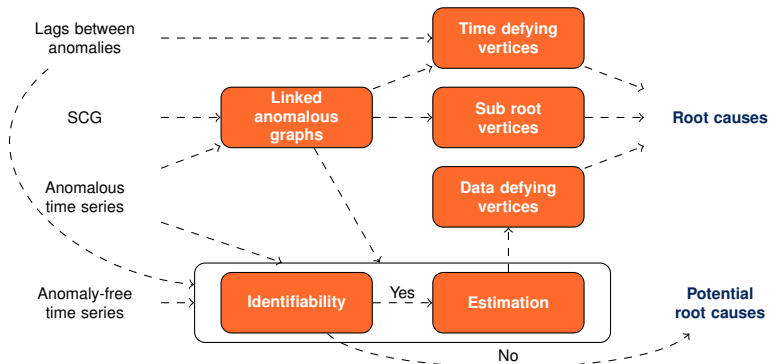
EasyRCA



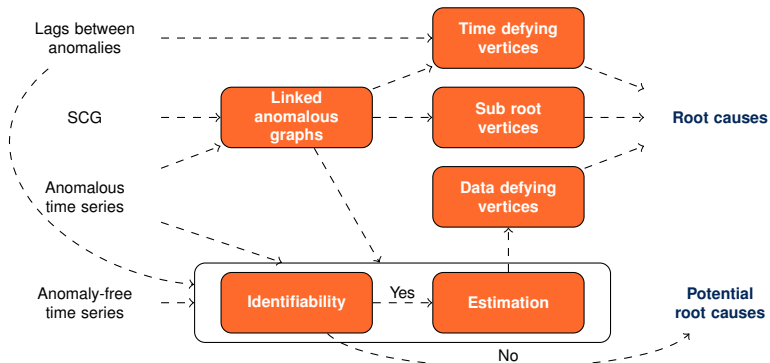
EasyRCA



EasyRCA



EasyRCA



Theorem

EasyRCA is able to detect all root causes if all assumptions are satisfied and if all direct effects are identifiable.

A heuristic for action recommendation

- 1 Fix root causes with high certainty
- 2 Fix root causes that provoked a disruption in the system
- 3 Fix other root causes

A heuristic for action recommendation

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Which means

- 1 Fix sub root vertices and time defying vertices
- 2 Fix data defying vertices with structural interventions
- 3 Fix data defying vertices with parametric interventions

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Distinguishing between parametric and structural interventions:

Given that $\alpha_{Y_t, X_t - \gamma_{X,Y}} \neq \bar{\alpha}_{Y_t, X_t - \gamma_{X,Y}}$

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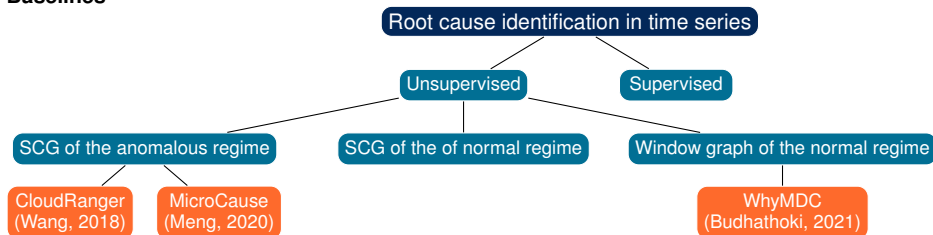
- there is a structural intervention on Y ,
- otherwise, there is a parametric intervention on Y .

Outline

- 1 Overview
 - IT monitoring systems
 - Causal graphs and abstractions
 - Root causes and anomalies
- 2 Root cause analysis using causal graphs
 - A root cause analysis framework using causal discovery
 - New root cause analysis framework
- 3 Experiments
- 4 Conclusion

Experimental set up

Baselines



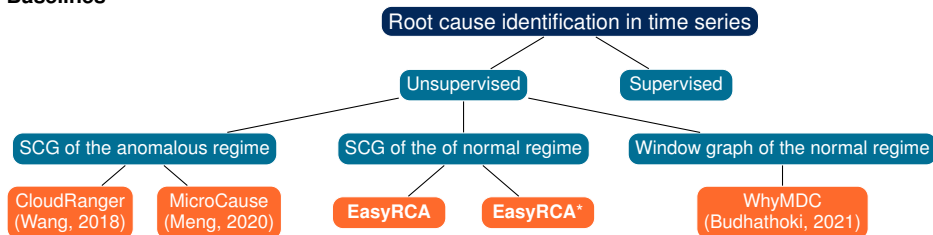
P. Wang et al. *CloudRanger: Root cause identification for cloud native systems*. IEEE/ACM CCGRID, 2018.

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Experimental set up

Baselines



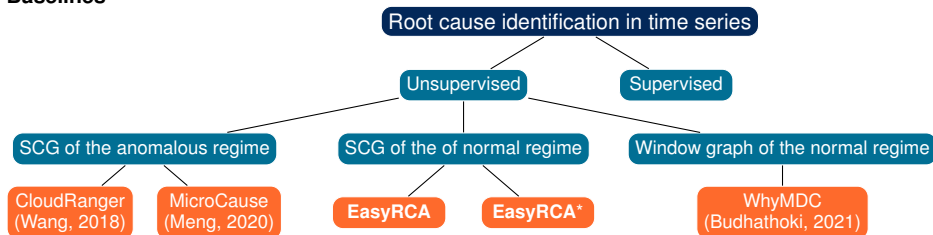
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Baselines



Estimation

- Grubb-test
- t-test
- ...

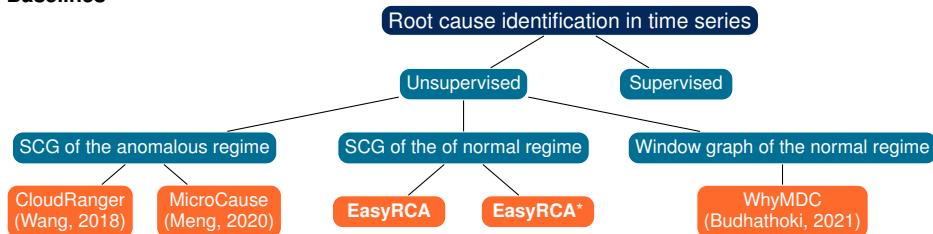
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Experimental set up

Baselines



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- t-test
- ...

Hyper-parameters

- $\gamma_{max} = 3$
- Significance level = 0.05
- ...

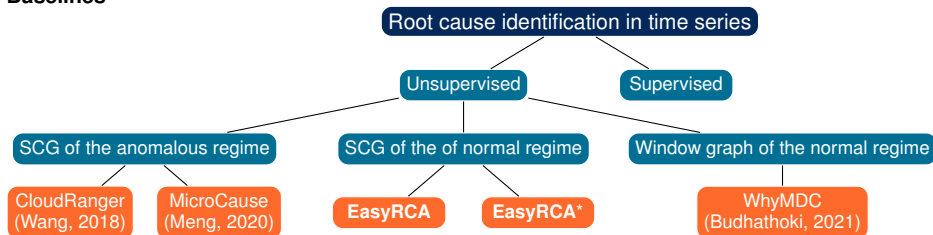
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Y. Meng et al. *Localizing failure root causes in a microservice through causality inference*. IWQoS, 2020.

K. Budhathoki et al. *Why did the distribution change?*. AISTATS, 2021.

Experimental set up

Baselines



Estimation

- Grubb-test
- t-test
- ...

Hyper-parameters

- $\gamma_{max} = 3$
- Significance level = 0.05
- ...

Evaluation measure

- F1-score on root causes that are neither sub root vertices nor time defying vertices.

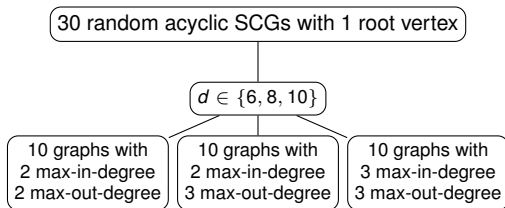
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Simulated data

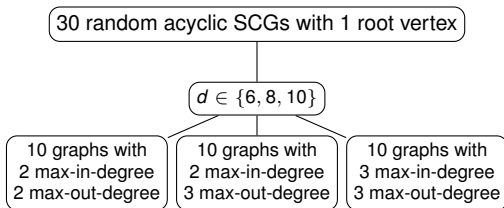
Generating SCGs:



- 1 intervention on 1 root vertex and 1 intervention on random non-root vertex

Simulated data

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Generating time series :

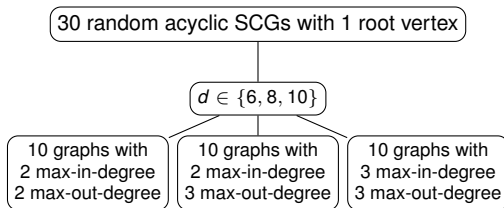
- 10 different datasets for each graph

$$Y_t = aY_{t-1} + \sum_{X \in \text{Parents}(Y)} bX_{t-1} + 0.1\xi_t,$$

- $a, b \sim U(-1, 0.1) \cup U(0.1, 1)$ and $\xi_t \sim N(0, 1)$
- Anomaly size $\in \{100, 2000\}$

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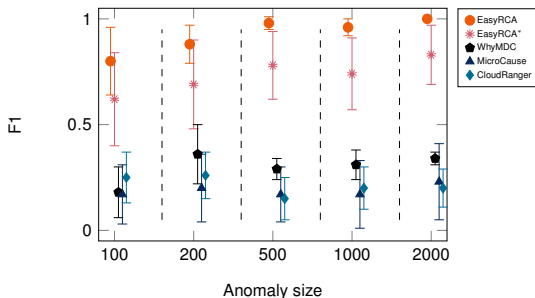
- $a, b \sim U(-1, 0.1) \cup U(0.1, 1)$ and $\xi_t \sim N(0, 1)$
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Generating interventions:

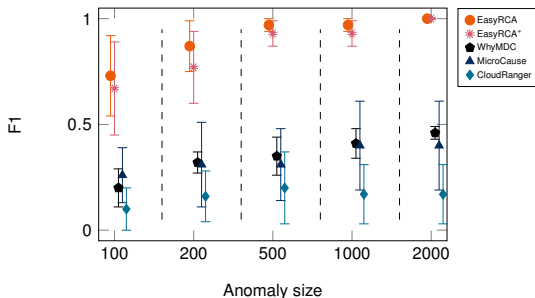
- In case of parametric interventions: $Y_t = aY_{t-1} + \sum_{X \in \text{Parents}(Y)} cX_{t-1} + 0.1\xi_t$
- In case of structural interventions: $Y_t \sim \text{Exp}(2)$

Simulated data: results

Parametric interventions

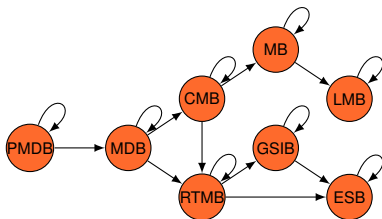


Structural interventions



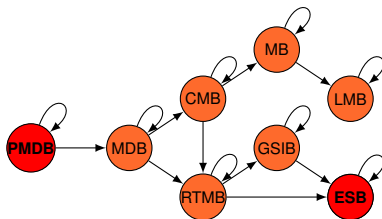
Real IT data

SCG of a message ingestion system



Real IT data

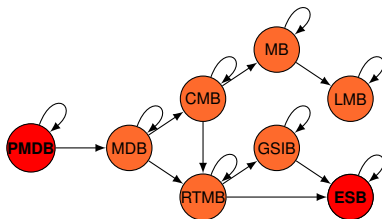
SCG of a message ingestion system



According to our assumption and to the expert knowledge, we expect PMDB and ESB to be the root causes.

Real IT data

SCG of a message ingestion system

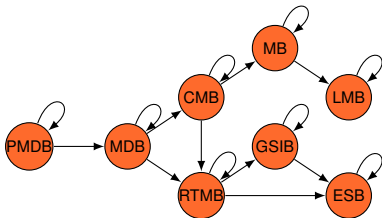


According to our assumption and to the expert knowledge, we expect PMDB and ESB to be the root causes.

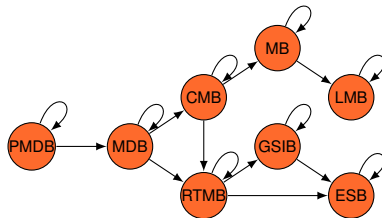
- 1 dataset
- Anomaly size = 230

Real IT data: results

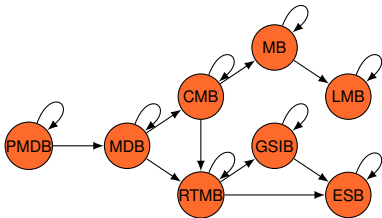
EasyRCA



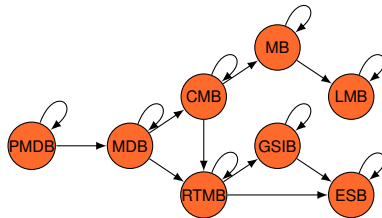
EasyRCA*



CloudRanger

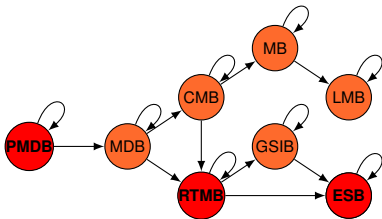


MicroCause

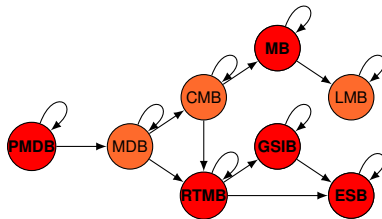


Real IT data: results

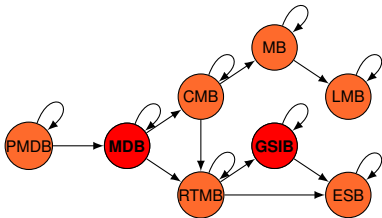
EasyRCA



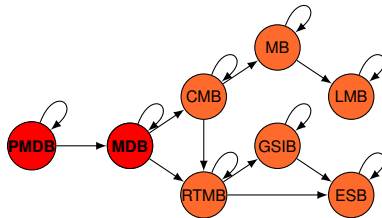
EasyRCA*



CloudRanger



MicroCause



Outline

- 1 Overview
 - IT monitoring systems
 - Causal graphs and abstractions
 - Root causes and anomalies
- 2 Root cause analysis using causal graphs
 - A root cause analysis framework using causal discovery
 - New root cause analysis framework
- 3 Experiments
- 4 Conclusion

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 - Small size of anomalous data
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 - Strong assumptions
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 - EasyRCA highly outperform other graph-based method with less computation time
- Identifiability of direct effects
 - Always possible for ASCGL
 - Complete result for SCGs

Main limitations and main perspectives

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- Limitation 2: Sometimes there are no experts to collaborate with a causal discovery algorithm
- Limitation 3: Interventions should directly affect the structural causal model $\implies$ cannot handle intervention on the noise

Other perspectives

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- A complete criterion to get all adjustment sets (**In progress**)

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- Avoid recommending unnecessary actions

Relevant papers

- C. K. Assaad, I. Ez-zejjari, and L. Zan. Root cause identification for collective anomalies in time series given an acyclic summary causal graph with loops. AISTATS, 2023.
- S. Ferreira and C. K. Assaad. Challenges of root Cause Identification for Collective Anomalies in Time Series given a Summary Causal Graph. Colloquium on When Causal Inference meets Statistical Analysis, 2023.
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Thank you !